

Bradley Zykoski

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RESEARCH INTERESTS Topology, geometry, and dynamics: Teichmüller theory, translation surfaces, triangulations

EMPLOYMENT *Northwestern University*, Evanston, Illinois
RTG Postdoctoral Fellow, September 2023 – August 2026 (projected)

EDUCATION *University of Michigan*, Ann Arbor, Michigan
Ph.D. in Mathematics, September 2017 – April 2023
Advisor: Alex Wright, alexmw@umich.edu

University of Michigan, Ann Arbor, Michigan
M.S. in Mathematics, May 2019

University of Virginia, Charlottesville, Virginia
B.A. with Highest Distinction in Mathematics, May 2017
Minor: Philosophy

RESEARCH With Vincent Delecroix and Kai Fu, *Computing the multi-scale boundary of real linear subvarieties in the moduli space of Abelian differentials*.
In preparation.

With C. Buyalos, J. Thadani, X. Wang, and M. Zshornack, *Some Families of Half-Relation Numbers and their Accumulation Points*.
Preprint available at <https://arxiv.org/abs/2511.10422>

With Sam Freedman, *Convex iso-Delaunay regions in strata of translation surfaces*.
Preprint available at <https://arxiv.org/abs/2509.19550>

The L^∞ -isodelaunay decomposition of strata of abelian differentials.
Preprint available at <https://arxiv.org/abs/2206.04143>.

With Anne Larsen and Chaya Norton, *Strongly obtuse rational lattice triangles*.
Transactions of the AMS, Vol. 374 (2021), pp. 7119-7142.

With M.I. Bueno, F.M. Dopico, J. Pérez, and R. Saavedra, *A simplified approach to Fiedler-like pencils via block minimal bases pencils*.
Linear Algebra and its Applications, Vol. 547 (2018), pp. 45-104.

With Greg Muller and Jenna Rajchgot, *Lower bound cluster algebras: presentations, Cohen-Macaulayness, and normality*. Algebraic Combinatorics, Vol. 1 (2018), pp. 95-114.

TEACHING**Northwestern University:**

Fall 2025	Math 354-0	Chaotic Dynamical Systems	Primary Instructor
Winter 2025	Math 105-8	First-Year Writing Seminar: The Unreasonable Effectiveness of Mathematics	Primary Instructor
Fall 2024	Math 220-1	Single-Variable Differential Calculus	Primary Instructor
Winter 2024	Math 230-1	Multivariable Differential Calculus	Primary Instructor
Fall 2023	Math 220-1	Single-Variable Differential Calculus	Primary Instructor

University of Michigan:

Winter 2023	Math 216	Introduction to Differential Equations	Lab Instructor
Fall 2022	Math 115	Calculus I	Primary Instructor
Fall 2021	Math 216	Introduction to Differential Equations	Lab Instructor
Summer 2021	Math 115	Calculus I	Instructor/Co-coordinator
Winter 2021	Math 115	Calculus I	Course Co-coordinator
Fall 2020	Math 115	Calculus I	Primary Instructor
Fall 2019	Math 116	Calculus II	Primary Instructor
Winter 2019	Math 116	Calculus II	Primary Instructor
Fall 2018	Math 115	Calculus I	Primary Instructor
Winter 2018	Math 115	Calculus I	Primary Instructor
Fall 2017	Math 105	Data, Functions, and Graphs	Primary Instructor

University of Virginia:

Spring 2017	Math 4720	Introduction to Differential Geometry	Grader
Fall 2016	Math 3340	Complex Variables with Applications	Grader
Spring 2015	Math 3310	Basic Real Analysis	Grader
2015-2017, Mathematics Tutor, University of Virginia Math Tutoring Center			

AWARDS

<i>Project NeXT Fellow</i>	
Mathematical Association of America	2025
<i>President's Challenge</i>	
University of Michigan Mathematics Department	2020
<i>E.J. McShane Prize</i>	
University of Virginia Mathematics Department	2017
<i>Honorable Mention</i>	
Goldwater Foundation	2016
<i>E.E. Floyd Prize</i>	
University of Virginia Mathematics Department	2015
<i>Phi Eta Sigma Honor Society</i>	Member

SERVICE

<i>Project Supervisor</i>	
Northwestern University RTG REU Program	Summer 2025
<i>Co-organizer</i>	
Dynamical Systems Seminar at Northwestern University	Fall 2024 – Spring 2025

SELECTED TALKS & POSTERS	<i>Organizer</i>	
	Directed Reading Program at Northwestern University	Fall 2023 – Present
	<i>Co-organizer</i>	
	NU Trends in Ergodic Theory 2024 Conference	July/August 2024
	<i>Volunteer and Facilitator</i>	
	Evanston Math Circle	January 2025 – Present
	<i>Helper and facilitator</i>	
	Wayne County Math Teachers' Circle	2020 – 2023
	<i>Mentor</i>	
	Directed Reading Program with a student studying triangle groups	Winter 2022
	<i>Mentor</i>	
	LOG(M) project with a small team of undergraduates	
	Project title: Curves in surfaces and mapping class group	Fall 2019
	<i>Co-organizer</i>	
	UM Graduate Student Topology Seminar	2019-2020
	<i>Co-organizer</i>	
	UM Graduate Student Dynamics Seminar	2019-2020
	<i>Contractibility of L^2 Isodelaunay Regions of Translation Surfaces</i>	
	Midwest Summer School in Geometry, Topology, and Dynamics	June 2025
	<i>A polytopal decomposition of strata of translation surfaces</i>	
	Geometry and Topology Seminar at the University of Chicago	November 2023
	AMS Sectional Meeting at the University at Buffalo	September 2023
	Topology/Geometric Group Theory Seminar at Ohio State University	February 2023
	Geometry Seminar at Indiana University	January 2023
	Geometry & Topology Seminar at the University of Warwick	November 2022
	Geometry & Topology Seminar at the University of Bristol	November 2022
	Geometry & Topology Seminar at the University of Glasgow	November 2022
	Dynamics Seminar at Northwestern University	October 2022
	Topology Seminar at the University of Notre Dame	October 2021
	Geometry Seminar at the University of Michigan	October 2021
	GSCAGT at Temple University	May 2021
	GSTGC at Indiana University	April 2021
	<i>Strongly obtuse rational lattice triangles</i>	
	Group Action Seminar at UC San Diego	February 2024
	<i>Principal bundles</i>	
	1-week mini-course at the University of Michigan	June 2020
	<i>Translation surfaces</i> (joint with Mark Greenfield)	
	1-week mini-course at the University of Michigan	August 2019
	<i>Pre-workshop learning group on Higgs bundles</i>	
	1-week mini-course at the University of Michigan	June 2018

<i>A unified approach to Fiedler-like pencils</i>	
SUMS Conference at JMU (poster)	September 2016
Young Mathematicians Conference at OSU	August 2016
Summer Undergraduate and Graduate Research Colloquium at UCSB (poster)	August 2016
<i>Lower bound cluster algebras and Stanley-Reisner complexes</i>	
Student Research Talks at GMU	October 2015
SUMS Conference at JMU	September 2015
Geometry Labs United Conference at UIUC (poster)	August 2015

**CONFERENCES
& WORKSHOPS
ATTENDED**

- MAA MathFest
Sacramento, August 2025
- NU Trends in Ergodic Theory and Ergodic Theory
Northwestern University, July/August 2024
- New Frontiers in Parabolic Dynamics and Renormalization
University of Bologna, June 2024
- AMS sectional meeting
University at Buffalo, September 2023
- Ergodic geometry, number theory and Margulis legacy: the next generation
University of Chicago, June 2022
- Ergodic Theory and its Connections: A Conference in Honor of the Legacy of Michael Boshernitzan
Rice University, May 2022
- Geometry, Groups, and Dynamics Workshop
IUPUI, April 2022
- Graduate Student Topology and Geometry Conference
Georgia Institute of Technology (hybrid format), April 2022
- A Week at Infinity
Online Conference, March 2022
- CMI-HIMR Dynamics and Geometry Online Summer School
Clay Mathematics Institute, Heilbronn Institute for Mathematical Research (virtual due to COVID-19), June 2021
- Graduate Student Conference in Algebra, Geometry, and Topology
Temple University (virtual due to COVID-19), May 2021
- Graduate Student Topology and Geometry Conference
Indiana University (virtual due to COVID-19), April 2021
- Winter School on Dynamics: Topology and Numbers
Hausdorff Research Institute for Mathematics, January 2020
- Flat Surfaces and Dynamics on Moduli Space, II
Casa Matemática Oaxaca, May 2019
- Graduate School on Geometry of Teichmüller Spaces
Simons Center for Geometry and Physics, April 2019
- Graduate Student Topology and Geometry Conference
University of Illinois at Urbana-Champaign, March 2019
- Junior Topology and Geometry in the Midwest
University of Wisconsin, Madison, October 2018

- Workshop on Geometry of Teichmüller Space
The Fields Institute, August 2018
- RTG Workshop on the Geometry and Physics of Higgs Bundles III
University of Illinois at Chicago, June 2018
- Graduate Student Topology and Geometry Conference
University of Illinois at Chicago, April 2018
- Shenandoah Undergraduate Mathematics and Statistics Conference
James Madison University, September 2016
- Young Mathematicians Conference
The Ohio State University, August 2016
- Shenandoah Undergraduate Mathematics and Statistics Conference
James Madison University, September 2015
- Geometry Labs United Conference
University of Illinois at Urbana-Champaign, August 2015